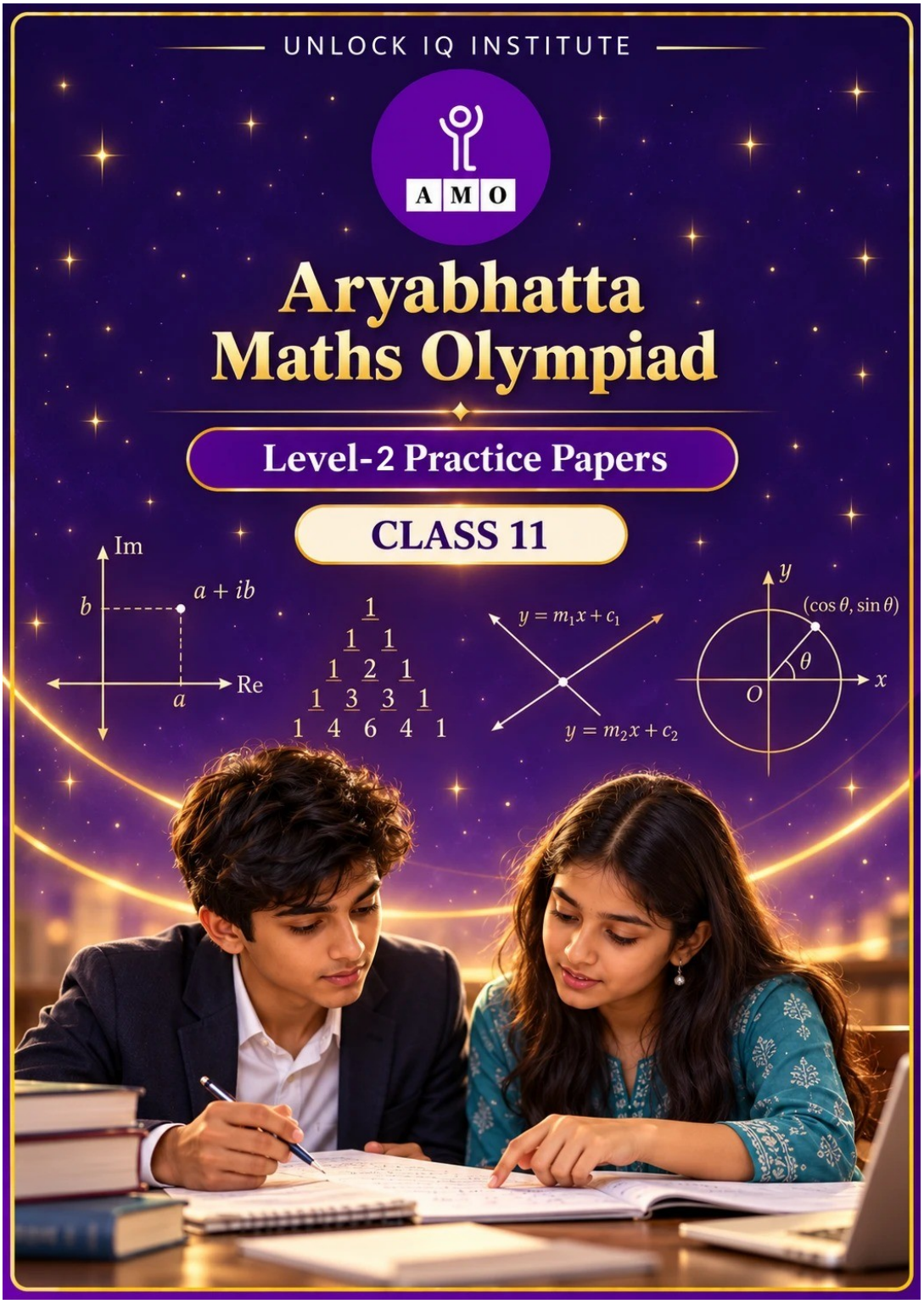


Set 1 · Practice Papers 1–5 · with answer keys & detailed solutions

Name: _____	**Class & Section:** _____
Roll No.: _____	**School:** _____
City: _____	**Date:** _____

Published by Unlock IQ Institute Pvt. Ltd.
Register for Olympiads: www.aryabhattamathsolympiad.com/student-registration

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-1

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. What is the value of $(1 + i)^{10} - (1 - i)^{10}$?

(A) 0 **(B)** 64 **(C)** 64i **(D)** −64i

Q2. A complex number z satisfies $z + |z| = 2 + 8i$. What is $|z|$?

(A) 15 **(B)** 8 **(C)** 17 **(D)** √68

Q3. Let $z = 1 + 2i$. What is the value of $z^3 - 2z^2 + 5z + 7$?

(A) 7 + 4i **(B)** 0 **(C)** −4 + 8i **(D)** 7

Part 2 — Algebra (Syllabus Chapter 4)

Q4. The sum of the first n terms of an AP is $3n^2 + 5n$ for every n. What is its 10th term?

(A) 350 **(B)** 62 **(C)** 59 **(D)** 65

Q5. In a geometric progression the 3rd term is 12 and the 6th term is −96. What is the sum of its first 8 terms?

(A) −255 **(B)** 765 **(C)** 255 **(D)** −765

Q6. Positive real numbers a and b satisfy $a + b = 10$. What is the greatest possible value of ab^2?

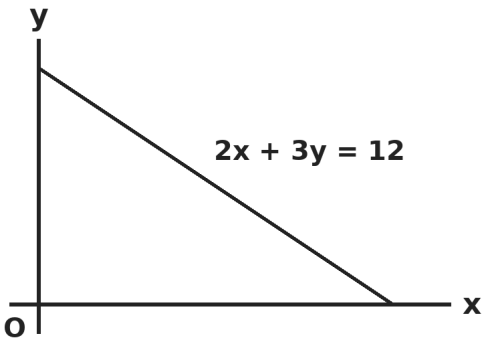
(A) 125 **(B)** 625/4 **(C)** 500/3 **(D)** 4000/27

Q7. What is the term independent of x in the expansion of $(x^2 - 2/x)^9$?

(A) −5376 **(B)** 672 **(C)** 5376 **(D)** −672

Q8. What is the value of $1\cdot2 + 2\cdot3 + 3\cdot4 + \ldots + 20\cdot21$?

(A) 3080 **(B)** 2870 **(C)** 3010 **(D)** 3290



Q9. How many points (x, y) with x and y both non-negative integers satisfy $2x + 3y \le 12$?

(A) 19 **(B)** 16 **(C)** 25 **(D)** 21

Part 3 — Geometry (Syllabus Chapter 5)

Q10. What is the mirror image of the point (1, 2) in the line $x + y = 7$?

(A) (3, 4) **(B)** (5, 6) **(C)** (6, 5) **(D)** (4, 3)

Q11. For which values of k are the lines $kx + 2y = 5$ and $3x + (k - 1)y = 7$ parallel? Give the sum of all such k.

(A) 5 **(B)** −6 **(C)** 3 **(D)** 1

Q12. What is the distance between the parallel lines $3x + 4y = 7$ and $6x + 8y + 11 = 0$?

(A) 5/2 **(B)** 18/5 **(C)** 1/2 **(D)** 3/10

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of $\cos 20° \times \cos 40° \times \cos 80°$?

(A) 1/8 **(B)** 1/4 **(C)** $\sqrt{3}/8$ **(D)** 1/16

Q14. If $\sin A = 1/3$, what is the value of $\sin 3A$?

(A) 1 **(B)** 25/27 **(C)** 1/27 **(D)** 23/27

Q15. What is the value of $\sin(-1110°) + \cos 1140° + \tan(-1035°)$?

(A) 0 **(B)** 2 **(C)** −1 **(D)** 1

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. All arrangements of the letters of MOTHER are listed in dictionary order. In which position does MOTHER itself appear?

(A) 308 **(B)** 240 **(C)** 309 **(D)** 313

Q17. A committee of 5 is chosen from 6 boys and 4 girls. In how many ways can it be chosen if it must have at least 3 girls?

(A) 60 **(B)** 126 **(C)** 66 **(D)** 186

Q18. A fair die is thrown 4 times. What is the probability that exactly two different numbers appear among the four results?

(A) 35/216 **(B)** 5/72 **(C)** 25/216 **(D)** 7/36

Q19. What is the standard deviation of the ten numbers 2, 4, 6, … , 20?

(A) 33 **(B)** √(33/4) **(C)** 11 **(D)** √33

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. Kavya borrows ₹21,000 at 10% per annum compound interest and repays it in two equal yearly instalments, the first one year after the loan. What is each instalment?

(A) ₹11,550 **(B)** ₹12,100 **(C)** ₹10,500 **(D)** ₹12,705

Q21. A manufacturer sells goods to a wholesaler for ₹40,000, the wholesaler sells them to a retailer for ₹50,000, and the retailer sells to Rohan for ₹60,000. GST is 12% at each sale. After input tax credit, how much GST in all do the wholesaler and the retailer pay to the government?

(A) ₹13,200 **(B)** ₹7,200 **(C)** ₹2,400 **(D)** ₹1,200

Q22. Shop A gives successive discounts of 25% and 10% on the marked price and then adds 12% GST. Shop B, with the same marked price, gives one single discount and then adds 5% GST. Both final bills are equal. What single discount does Shop B give?

(A) 35% **(B)** 28% **(C)** 32.5% **(D)** 24.4%

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. Ishaan picks distinct numbers from 1, 2, 3, … , 20. What is the least number he must pick to be sure that two of his numbers add up to 21?

(A) 10 **(B)** 11 **(C)** 12 **(D)** 21

Q24. Six straight lines are drawn in a plane so that no two are parallel and no three pass through one point. Into how many regions do they divide the plane?

(A) 22 **(B)** 21 **(C)** 16 **(D)** 64

Q25. For which statement P(n) does "P(k) true implies P(k + 1) true" hold for every k, even though P(n) is false for every natural number n?

(A) $n^2 + n$ is even **(B)** $2^n > n$

(C) $n^2 + n + 1$ is even **(D)** $3^n - 1$ is even

Part 8 — Mixed Problems (each joins two chapters)

Q26. A sequence has $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for $n \geq 1$. What is $a_1 + a_2 + \ldots + a_{10}$?

(A) 2046 **(B)** 2036 **(C)** 1023 **(D)** 2026

Q27. What is the value of $1 \cdot C(10, 1) + 2 \cdot C(10, 2) + 3 \cdot C(10, 3) + \ldots + 10 \cdot C(10, 10)$?

(A) 5120 **(B)** 1024 **(C)** 10240 **(D)** 2560

Q28. What is the value of $1 - C(8, 2) + C(8, 4) - C(8, 6) + C(8, 8)$?

(A) 0 **(B)** 128 **(C)** −16 **(D)** 16

Q29. What is the acute angle between the lines $y = (2 - \sqrt{3})x$ and $y = (2 + \sqrt{3})x$?

(A) 30° **(B)** 60° **(C)** 45° **(D)** 75°

Q30. A selection (possibly empty) is made from the numbers 1, 2, 3, … , 10 so that no two chosen numbers are consecutive. How many such selections are there?

(A) 89 **(B)** 233 **(C)** 144 **(D)** 512

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 1

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 1

Quick answers

1 **C**	6 **D**	11 **D**	16 **C**	21 **C**	26 **B**
2 **C**	7 **C**	12 **A**	17 **C**	22 **B**	27 **A**
3 **D**	8 **A**	13 **A**	18 **A**	23 **B**	28 **D**
4 **B**	9 **A**	14 **D**	19 **D**	24 **A**	29 **B**
5 **A**	10 **B**	15 **D**	20 **B**	25 **C**	30 **C**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(C)** 64i	***Hint:*** *Square first: $(1 + i)^2 = 2i$ and $(1 - i)^2 = -2i$.* $(1 + i)^{10} = (2i)^5 = 32i^5 = 32i$. $(1 - i)^{10} = (-2i)^5 = -32i^5 = -32i$. The difference is $32i - (-32i) = 64i$.
2	Part 1 · Ch1 · Hard	**(C)** 17	***Hint:*** *Write $z = a + 8i$, because $\|z\|$ is real and so the imaginary part of z must be 8.* The imaginary part of z is 8, so $z = a + 8i$ and $a + \sqrt{(a^2 + 64)} = 2$. Then $\sqrt{(a^2 + 64)} = 2 - a$; squaring gives $a^2 + 64 = 4 - 4a + a^2$, so $a = -15$. $\|z\| = 2 - a = 17$.
3	Part 1 · Ch1 · Hard	**(D)** 7	***Hint:*** *Check first whether z is a root of $z^2 - 2z + 5 = 0$.* $z^2 = (1 + 2i)^2 = -3 + 4i$, so $z^2 - 2z + 5 = -3 + 4i - 2 - 4i + 5 = 0$. Hence $z^3 - 2z^2 + 5z + 7 = z(z^2 - 2z + 5) + 7 = 7$.
4	Part 2 · Ch4 · Hard	**(B)** 62	***Hint:*** *The nth term is $S_n - S_{n-1}$.* $T_{10} = S_{10} - S_9 = (300 + 50) - (243 + 45) = 350 - 288 = 62$. (In general $T_n = 6n + 2$.)
5	Part 2 · Ch4 · Hard	**(A)** −255	***Hint:*** *Divide the 6th term by the 3rd term to get r^3.* $r^3 = -96 \div 12 = -8$, so $r = -2$ and $a = 12 \div 4 = 3$. $S_8 = 3(1 - (-2)^8)/(1 - (-2)) = 3(1 - 256)/3 = -255$.
6	Part 2 · Ch4 · Hard	**(D)** 4000/27	***Hint:*** *Apply AM ≥ GM to the three numbers a, $b/2$ and $b/2$.* $a + b/2 + b/2 = 10$, so by AM ≥ GM, $a \cdot (b/2) \cdot (b/2) \le (10/3)^3 = 1000/27$. Hence $ab^2 \le 4 \times 1000/27 = 4000/27$, with equality when $a = b/2$, that is $a = 10/3$, $b = 20/3$.
7	Part 2 · Ch4 · Hard	**(C)** 5376	***Hint:*** *The general term has x raised to $18 - 3r$.* $T_{r+1} = C(9, r)(x^2)^{9-r}(-2/x)^r = C(9, r)(-2)^r x^{18-3r}$. For x^0, $r = 6$: $C(9, 6) \times (-2)^6 = 84 \times 64 = 5376$.
8	Part 2 · Ch4 · Hard	**(A)** 3080	***Hint:*** *Write $k(k + 1) = k^2 + k$ and use the formulas for Σk^2 and Σk.* $\Sigma k(k + 1) = \Sigma k^2 + \Sigma k = 20 \cdot 21 \cdot 41/6 + 20 \cdot 21/2 = 2870 + 210 = 3080$. (Also $n(n + 1)(n + 2)/3 = 20 \cdot 21 \cdot 22/3 = 3080$.)
9	Part 2 · Ch4 · Hard	**(A)** 19	***Hint:*** *Fix $y = 0, 1, 2, 3, 4$ and count the allowed x each time.* $y = 0$: $x = 0$ to 6 (7). $y = 1$: $2x \le 9$, $x = 0$ to 4 (5). $y = 2$: $x = 0$ to 3 (4). $y = 3$: $x = 0, 1$ (2). $y = 4$: $x = 0$ (1). Total $7 + 5 + 4 + 2 + 1 = 19$.
10	Part 3 · Ch5 · Hard	**(B)** (5, 6)	***Hint:*** *First find the foot of the perpendicular; the image is twice as far.* The perpendicular from (1, 2) has direction (1, 1). Foot: $(1 + t, 2 + t)$ with $3 + 2t = 7$, so $t = 2$ and the foot is (3, 4). The image is $(1 + 4, 2 + 4) = (5, 6)$.

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(D)** 1	***Hint:*** *Parallel means the coefficient ratios of x and y match, but the lines are not the same line.* $k/3 = 2/(k-1)$ gives $k^2 - k - 6 = 0$, so $k = 3$ or $k = -2$. For $k = 3$ the lines are $3x + 2y = 5$ and $3x + 2y = 7$; for $k = -2$ they are $-2x + 2y = 5$ and $3x - 3y = 7$. Neither pair coincides, so both work. Sum = 1.
12	Part 3 · Ch5 · Hard	**(A)** 5/2	***Hint:*** *Make the x and y coefficients the same in both equations first.* Divide the second line by 2: $3x + 4y + 5.5 = 0$. The first is $3x + 4y - 7 = 0$. Distance $= \lvert -7 - 5.5 \rvert \div \sqrt{9 + 16} = 12.5 \div 5 = 5/2$. (18/5 forgets to halve the second equation.)
13	Part 4 · Ch7 · Hard	**(A)** 1/8	***Hint:*** *Multiply and divide by sin 20° and use 2 sin A cos A = sin 2A three times.* Multiply by $8 \sin 20°$: $8 \sin 20° \cos 20° \cos 40° \cos 80° = 4 \sin 40° \cos 40° \cos 80° = 2 \sin 80° \cos 80° = \sin 160° = \sin 20°$. So the product is $\sin 20° \div (8 \sin 20°) = 1/8$.
14	Part 4 · Ch7 · Hard	**(D)** 23/27	***Hint:*** *$\sin 3A = 3 \sin A - 4 \sin^3 A$.* $\sin 3A = 3(1/3) - 4(1/3)^3 = 1 - 4/27 = 23/27$. (Taking $\sin 3A = 3 \sin A$ gives the wrong answer 1.)
15	Part 4 · Ch7 · Hard	**(D)** 1	***Hint:*** *Remove complete turns of 360°, and use $\sin(-\theta) = -\sin\theta$ and $\tan(-\theta) = -\tan\theta$.* $1110° = 3 \times 360° + 30°$, so $\sin(-1110°) = -\sin 30° = -1/2$. $1140° = 3 \times 360° + 60°$, so $\cos 1140° = 1/2$. $1035° = 3 \times 360° - 45°$, so $\tan(-1035°) = -\tan(-45°) = \tan 45° = 1$. Total: $-1/2 + 1/2 + 1 = 1$.
16	Part 5 · Ch8 · Hard	**(C)** 309	***Hint:*** *Count the words that start with a letter earlier than M, then fix M and repeat.* Letters in order: E H M O R T. Starting E or H: $2 \times 5! = 240$. MO… with second letter E or H: $2 \times 4! = 48$. MOT… with third letter E, H or R: $3 \times 3! = 18$. MOTH… with E next: $1 \times 2! = 2$. Then MOTHER. Rank $= 240 + 48 + 18 + 2 + 1 = 309$.
17	Part 5 · Ch8 · Hard	**(C)** 66	***Hint:*** *Split into the cases of exactly 3 girls and exactly 4 girls.* 3 girls and 2 boys: $C(4, 3) \times C(6, 2) = 4 \times 15 = 60$. 4 girls and 1 boy: $C(4, 4) \times C(6, 1) = 6$. Total 66.
18	Part 5 · Ch8 · Hard	**(A)** 35/216	***Hint:*** *Choose the two numbers, then count the ways to fill 4 throws using both of them.* Choose the two numbers: $C(6, 2) = 15$. Fill 4 throws with only these two, using both: $2^4 - 2 = 14$. Favourable $= 15 \times 14 = 210$, so $P = 210/1296 = 35/216$.
19	Part 5 · Ch8 · Hard	**(D)** √33	***Hint:*** *The numbers are $2 \times (1, 2, \ldots, 10)$. The variance of $1, 2, \ldots, n$ is $(n^2 - 1)/12$.* Variance of 1 to 10 $= (100 - 1)/12 = 33/4$. Multiplying every value by 2 multiplies the variance by 4: 33. SD = √33.
20	Part 6 · Ch9 · Hard	**(B)** ₹12,100	***Hint:*** *The present values of the two instalments must add up to ₹21,000.* If each instalment is x, then $x/1.1 + x/1.21 = 21{,}000$, so $x(1.1 + 1)/1.21 = 21{,}000$ and $x = 21{,}000 \times 1.21 \div 2.1 =$ ₹12,100.
21	Part 6 · Ch9 · Hard	**(C)** ₹2,400	***Hint:*** *Each trader pays GST on the value they add: output GST minus input GST.* Wholesaler: 12% of 50,000 − 12% of 40,000 = 6,000 − 4,800 = ₹1,200. Retailer: 7,200 − 6,000 = ₹1,200. Together ₹2,400, which is 12% of the value they added (₹20,000). (₹13,200 adds their full output GST.)

Q	Part · level	Answer	Why?
22	Part 6 · Ch9 · Hard	**(B)** 28%	***Hint:*** *Write each final bill as a fraction of the marked price.* Shop A: 0.75 × 0.90 × 1.12 = 0.756 of the marked price. Shop B: (1 − d) × 1.05 = 0.756, so 1 − d = 0.72 and d = 28%. (32.5% is the combined discount of A before GST; 24.4% ignores B's GST.)
23	Part 7 · Ch10 · Hard	**(B)** 11	***Hint:*** *Group the numbers into pairs that add up to 21.* The pairs (1, 20), (2, 19), … , (10, 11) are 10 boxes. With 10 numbers he can take one from each pair and avoid a sum of 21. With 11 numbers, two must come from the same pair (pigeonhole), and they add up to 21.
24	Part 7 · Ch10 · Hard	**(A)** 22	***Hint:*** *The kth line crosses the earlier k − 1 lines and so adds k new regions.* Starting from 1 region, line k adds k regions. Total = 1 + (1 + 2 + 3 + 4 + 5 + 6) = 1 + 21 = 22. (In general $(n^2 + n + 2)/2$.)
25	Part 7 · Ch10 · Hard	**(C)** $n^2 + n + 1$ is even	***Hint:*** *Check the base case of each statement, then the step.* $(k + 1)^2 + (k + 1) + 1 = (k^2 + k + 1) + 2(k + 1)$, so if $k^2 + k + 1$ is even, so is the next one: the step works. But $n^2 + n = n(n + 1)$ is always even, so $n^2 + n + 1$ is always odd: the statement is false. The other three are true for all n.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(B)** 2036	***Hint:*** *Look at $a_n + 1$. Prove the pattern by induction.* $a_n + 1$ doubles each time: $a_{n+1} + 1 = 2(a_n + 1)$, and $a_1 + 1 = 2$, so $a_n = 2^n - 1$. Sum = $(2 + 4 + \ldots + 2^{10}) - 10 = 2046 - 10 = 2036$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(A)** 5120	***Hint:*** *$k \cdot C(10, k) = 10 \cdot C(9, k - 1)$.* $k \cdot C(10, k) = 10 \cdot C(9, k - 1)$. So the sum is $10 \times [C(9, 0) + C(9, 1) + \ldots + C(9, 9)] = 10 \times 2^9 = 5120$.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(D)** 16	***Hint:*** *These are the real terms in the binomial expansion of $(1 + i)^8$.* $(1 + i)^8 = \Sigma\, C(8, k) i^k$. The real terms are those with even k, with signs +, −, +, −, +. So the value is the real part of $(1 + i)^8 = (2i)^4 = 16$.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(B)** 60°	***Hint:*** *Use $\tan\theta = \lvert (m_1 - m_2)/(1 + m_1 m_2) \rvert$.* $m_1 - m_2 = -2\sqrt{3}$ and $m_1 m_2 = 4 - 3 = 1$. $\tan\theta = 2\sqrt{3} \div 2 = \sqrt{3}$, so $\theta = 60°$. (The slopes are tan 15° and tan 75°, which also gives 60°.)
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(C)** 144	***Hint:*** *Let f_n be the count for the numbers 1 to n. Either n is left out, or n is in and n − 1 is left out.* $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 2$ (choose nothing, or choose 1) and $f_2 = 3$. Then 5, 8, 13, 21, 34, 55, 89, 144. So $f_{10} = 144$.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-2

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. For which real numbers a is $(a + 4i)/(1 + ai)$ a real number? Find the product of all such values of a.

(A) 4 **(B)** 0 **(C)** −4 **(D)** −2

Q2. Which of these is a square root of $-5 + 12i$?

(A) $3 + 2i$ **(B)** $2 - 3i$ **(C)** $-3 + 2i$ **(D)** $2 + 3i$

Q3. As n runs through all integers, how many different values can $i^n + i^{-n}$ take?

(A) 4 **(B)** 2 **(C)** 1 **(D)** 3

Part 2 — Algebra (Syllabus Chapter 4)

Q4. How many terms of the AP 24, 21, 18, ... must be added to get a sum of 78?

(A) 4 or 13 **(B)** 4 only **(C)** 13 only **(D)** 6 or 11

Q5. A ball is dropped from a height of 64 m. Each time it hits the ground it rises to 3/4 of the height it fell from. What total distance does it travel before coming to rest?

(A) 256 m **(B)** 384 m **(C)** 448 m **(D)** 512 m

Q6. What is the coefficient of x^4 in the expansion of $(2x - 3)^7$?

(A) 15120 **(B)** −15120 **(C)** −7560 **(D)** 560

Q7. What is the value of $11^3 + 12^3 + 13^3 + \ldots + 20^3$?

(A) 41075 **(B)** 44100 **(C)** 38025 **(D)** 41175

Q8. What is the least value of $4^x + 4^{1-x}$ for real x?

(A) 2 **(B)** 4 **(C)** 5 **(D)** 1

Q9. How many integers x satisfy $3x - 5 < 2x + 1 \le 4x + 9$?

(A) 10 **(B)** 9 **(C)** 11 **(D)** 12

Part 3 — Geometry (Syllabus Chapter 5)

Q10. A line passes through (4, 0) and (0, 3). What is the foot of the perpendicular from the origin to this line?

(A) (2, 3/2) **(B)** (48/25, 36/25)

(C) (36/25, 48/25) **(D)** (12/5, 0)

Q11. For what value of k do the lines $2x + y = 5$, $x - y = 1$ and $kx + 3y = 10$ pass through one point?

(A) 7/2 **(B)** 3 **(C)** 5/2 **(D)** 4

Q12. Two opposite sides of a square lie on the lines $x + y = 1$ and $x + y = 7$. What is the area of the square?

(A) 36 **(B)** 9 **(C)** 72 **(D)** 18

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of $\tan 15° + \cot 15°$?

(A) $2\sqrt{3}$ **(B)** 2 **(C)** 4 **(D)** $\sqrt{3} + 1$

Q14. In a triangle ABC, $\tan A = 1$ and $\tan B = 2$. What is $\tan A + \tan B + \tan C$?

(A) 3 **(B)** 6 **(C)** −3 **(D)** 0

Q15. An angle A between 45° and 90° satisfies $\sin A + \cos A = 7/5$. What is $\cos 2A$?

(A) −7/25 **(B)** 7/25 **(C)** 24/25 **(D)** −24/25

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. In how many arrangements of the letters of BANANA are the two letters N not next to each other?

(A) 60 **(B)** 20 **(C)** 40 **(D)** 50

Q17. How many four-digit numbers with all digits different are divisible by 5?

(A) 952 **(B)** 1008 **(C)** 504 **(D)** 896

Q18. A binomial random variable has mean 4 and variance 3. How many trials does it have?

(A) 16 **(B)** 12 **(C)** 7 **(D)** 4

Q19. Five observations have mean 10 and variance 4. Three of them are 9, 10 and 11. What is the larger of the other two?

(A) 12 **(B)** 14 **(C)** 15 **(D)** 13

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. Shalini invests ₹8,000. The bank pays compound interest at 10% in the first year, 20% in the second and 25% in the third. How much interest does she earn in three years?

(A) ₹4,400 **(B)** ₹5,200 **(C)** ₹13,200 **(D)** ₹4,800

Q21. A small unit has fixed costs of ₹60,000. Each item costs ₹150 to make and sells for ₹250. How many items must it sell to make a profit of ₹40,000?

(A) 600 **(B)** 400 **(C)** 667 **(D)** 1000

Q22. A boat goes 36 km downstream and returns the same 36 km upstream in 5 hours in all. In the time it takes to go 12 km downstream, it can go 8 km upstream. What is the speed of the stream?

(A) 6 km/h **(B)** 15 km/h **(C)** 3 km/h **(D)** 2 km/h

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the smallest number of integers that must be chosen from 1, 2, 3, … , 20 to be sure that one of the chosen numbers divides another?

(A) 10 **(B)** 21 **(C)** 16 **(D)** 11

Q24. Four letters are put into four addressed envelopes, one in each, so that no letter is in its own envelope. In how many ways can this be done?

(A) 6 **(B)** 9 **(C)** 11 **(D)** 23

Q25. What is the smallest natural number N such that $2^n > n^3$ for every $n \geq N$?

(A) 9 **(B)** 11 **(C)** 2 **(D)** 10

Part 8 — Mixed Problems (each joins two chapters)

Q26. What is the value of $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + 5 \cdot 5! + 6 \cdot 6!$?

(A) 5040 **(B)** 4319 **(C)** 5039 **(D)** 873

Q27. Three fair dice are thrown. What is the probability that the numbers add up to 10?

(A) 1/6 **(B)** 5/36 **(C)** 7/72 **(D)** 1/8

Q28. What is the value of $1 + i + i^2 + i^3 + \ldots + i^{100}$?

(A) 0 **(B)** 1 **(C)** i **(D)** 101

Q29. A line through the origin makes an angle of 15° with the positive x-axis. What is the perpendicular distance from the point (1, 0) to this line?

(A) $(\sqrt{6} + \sqrt{2})/4$ **(B)** $2 - \sqrt{3}$ **(C)** $(\sqrt{6} - \sqrt{2})/4$ **(D)** 1/2

Q30. Aarav climbs a staircase of 10 steps, taking either 1 or 2 steps at a time. In how many different ways can he reach the top?

(A) 55 **(B)** 89 **(C)** 144 **(D)** 100

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 2

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 2

Quick answers

1 **C**	6 **B**	11 **A**	16 **C**	21 **D**	26 **C**
2 **D**	7 **A**	12 **D**	17 **A**	22 **C**	27 **D**
3 **D**	8 **B**	13 **C**	18 **A**	23 **D**	28 **B**
4 **A**	9 **A**	14 **B**	19 **D**	24 **B**	29 **C**
5 **C**	10 **C**	15 **A**	20 **B**	25 **D**	30 **B**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(C)** −4	***Hint:*** *Multiply the top and bottom by the conjugate 1 − ai and set the imaginary part to 0.* $(a + 4i)(1 - ai) = a - a^2 i + 4i + 4a = 5a + (4 - a^2)i$, and the denominator $1 + a^2$ is real. The number is real when $4 - a^2 = 0$, so $a = 2$ or $a = -2$. Product = −4.
2	Part 1 · Ch1 · Hard	**(D)** 2 + 3i	***Hint:*** *Square each option, or solve $(a + bi)^2 = -5 + 12i$ with $a^2 - b^2 = -5$ and $ab = 6$.* $(2 + 3i)^2 = 4 - 9 + 12i = -5 + 12i$. The others give 5 + 12i, −5 − 12i and 5 − 12i.
3	Part 1 · Ch1 · Hard	**(D)** 3	***Hint:*** *The value depends only on n modulo 4.* $n \equiv 0$: $1 + 1 = 2$. $n \equiv 1$: $i + 1/i = i - i = 0$. $n \equiv 2$: $-1 - 1 = -2$. $n \equiv 3$: $-i + i = 0$. The values are 2, 0 and −2: three in all.
4	Part 2 · Ch4 · Hard	**(A)** 4 or 13	***Hint:*** *Set $S_n = 78$ and solve the quadratic in n. Check both roots.* $S_n = n[48 - 3(n - 1)]/2 = 78$ gives $3n^2 - 51n + 156 = 0$, so $n^2 - 17n + 52 = 0$, n = 4 or 13. Both are valid: 24 + 21 + 18 + 15 = 78, and the terms from the 5th to the 13th (12, 9, … , −12) add up to 0.
5	Part 2 · Ch4 · Hard	**(C)** 448 m	***Hint:*** *After the first fall, every rise is matched by an equal fall.* Total = $64 + 2(48 + 36 + 27 + \ldots) = 64 + 2 \times 48/(1 - 3/4) = 64 + 2 \times 192 = 448$ m.
6	Part 2 · Ch4 · Hard	**(B)** −15120	***Hint:*** *The x^4 term comes from $C(7, 4)(2x)^4(-3)^3$.* $C(7, 4) \times 2^4 \times (-3)^3 = 35 \times 16 \times (-27) = -15120$.
7	Part 2 · Ch4 · Hard	**(A)** 41075	***Hint:*** *Σk^3 from 1 to n is $[n(n + 1)/2]^2$.* $\Sigma_1^{20} k^3 = 210^2 = 44100$ and $\Sigma_1^{10} k^3 = 55^2 = 3025$. The required sum is 44100 − 3025 = 41075.
8	Part 2 · Ch4 · Hard	**(B)** 4	***Hint:*** *The product of the two terms is constant.* $4^x \times 4^{1-x} = 4$. By AM ≥ GM, $4^x + 4^{1-x} \geq 2\sqrt{4} = 4$, with equality when $4^x = 4^{1-x}$, that is $x = 1/2$.
9	Part 2 · Ch4 · Hard	**(A)** 10	***Hint:*** *Solve the two inequalities separately and take the overlap.* $3x - 5 < 2x + 1$ gives $x < 6$. $2x + 1 \leq 4x + 9$ gives $x \geq -4$. The integers are −4, −3, … , 5: 10 of them.
10	Part 3 · Ch5 · Hard	**(C)** (36/25, 48/25)	***Hint:*** *The line is 3x + 4y = 12. The perpendicular from O goes along the direction (3, 4).* The line is $x/4 + y/3 = 1$, that is $3x + 4y = 12$. The foot is t(3, 4) with $9t + 16t = 12$, so $t = 12/25$ and the foot is (36/25, 48/25). (Its distance from O is 12/5.)

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(A)** 7/2	***Hint:*** *Find where the first two lines meet.* Adding the first two: 3x = 6, so x = 2 and y = 1. The third line must pass through (2, 1): 2k + 3 = 10, so k = 7/2.
12	Part 3 · Ch5 · Hard	**(D)** 18	***Hint:*** *The side of the square is the distance between the two parallel lines.* Distance = $\|7 - 1\| \div \sqrt{(1 + 1)} = 6/\sqrt{2} = 3\sqrt{2}$. Area = $(3\sqrt{2})^2 = 18$. (36 forgets the √2.)
13	Part 4 · Ch7 · Hard	**(C)** 4	***Hint:*** *$\tan\theta + \cot\theta = 1/(\sin\theta\cos\theta) = 2/\sin 2\theta$.* $\tan\theta + \cot\theta = (\sin^2\theta + \cos^2\theta)/(\sin\theta\cos\theta) = 2/\sin 2\theta$. With θ = 15°: 2/sin 30° = 4. (Directly: (2 − √3) + (2 + √3) = 4.)
14	Part 4 · Ch7 · Hard	**(B)** 6	***Hint:*** *C = 180° − (A + B), so tan C = −tan(A + B).* $\tan(A + B) = (1 + 2)/(1 - 2) = -3$, so tan C = 3. Sum = 1 + 2 + 3 = 6. (It also equals the product 1 × 2 × 3, as it must when A + B + C = 180°.)
15	Part 4 · Ch7 · Hard	**(A)** −7/25	***Hint:*** *Square to get sin 2A, then find sin A − cos A.* Squaring: 1 + sin 2A = 49/25, so sin 2A = 24/25. $(\sin A - \cos A)^2 = 1 - 24/25 = 1/25$, and sin A > cos A because A > 45°, so sin A − cos A = 1/5. cos 2A = (cos A − sin A)(cos A + sin A) = (−1/5)(7/5) = −7/25.
16	Part 5 · Ch8 · Hard	**(C)** 40	***Hint:*** *Arrange B, A, A, A first, then put the two letters N into the gaps.* B, A, A, A can be arranged in 4!/3! = 4 ways. They leave 5 gaps, and the two letters N go into 2 different gaps in C(5, 2) = 10 ways. Total 4 × 10 = 40. (All arrangements: 60; with the letters N together: 20.)
17	Part 5 · Ch8 · Hard	**(A)** 952	***Hint:*** *The last digit is 0 or 5. Treat the two cases separately because 0 cannot be first.* Ending in 0: 9 × 8 × 7 = 504. Ending in 5: first digit 8 choices (not 0 or 5), then 8 and 7: 448. Total 952.
18	Part 5 · Ch8 · Hard	**(A)** 16	***Hint:*** *Mean = np and variance = npq.* npq/np = q = 3/4, so p = 1/4 and n = 4 ÷ 1/4 = 16.
19	Part 5 · Ch8 · Hard	**(D)** 13	***Hint:*** *Use the sum of the values and the sum of their squares.* Sum = 50, so the other two add up to 20. $\Sigma x^2 = 5(4 + 100) = 520$, so their squares add up to 520 − 302 = 218. With x + y = 20: 2xy = 400 − 218 = 182, xy = 91, so the values are 7 and 13.
20	Part 6 · Ch9 · Hard	**(B)** ₹5,200	***Hint:*** *Multiply the factors 1.1, 1.2 and 1.25.* Amount = 8,000 × 1.1 × 1.2 × 1.25 = ₹13,200. Interest = 13,200 − 8,000 = ₹5,200. (Simple addition of 55% gives ₹4,400.)
21	Part 6 · Ch9 · Hard	**(D)** 1000	***Hint:*** *Each item contributes ₹100 towards fixed costs and profit.* Contribution per item = 250 − 150 = ₹100. Needed: 60,000 + 40,000 = ₹1,00,000, so 1,000 items. (600 is only the break-even point.)
22	Part 6 · Ch9 · Hard	**(C)** 3 km/h	***Hint:*** *The downstream and upstream speeds are in the ratio 12 : 8 = 3 : 2.* Let the speeds be 3k and 2k. Then 36/(3k) + 36/(2k) = 5, so 12/k + 18/k = 5 and k = 6. Downstream 18 km/h, upstream 12 km/h, so the stream is (18 − 12)/2 = 3 km/h. (15 km/h is the boat's own speed.)
23	Part 7 · Ch10 · Hard	**(D)** 11	***Hint:*** *Write each number as $2^k \times$ (an odd number). How many odd parts are possible?* Every number is $2^k \times m$ with m odd, and m is one of 1, 3, 5, … , 19: ten classes. Two numbers in the same class divide one another. The ten numbers 11, 12, … , 20 show that 10 is not enough (none divides another). So 11.

Q	Part · level	Answer	Why?
24	Part 7 · Ch10 · Hard	**(B)** 9	***Hint:*** *Use inclusion–exclusion or list the cases.* $D_4 = 4!(1 - 1 + 1/2 - 1/6 + 1/24) = 24 - 24 + 12 - 4 + 1 = 9$.
25	Part 7 · Ch10 · Hard	**(D)** 10	***Hint:*** *Check n = 9 and n = 10, then show the step works for n ≥ 10.* $2^9 = 512 < 729 = 9^3$, but $2^{10} = 1024 > 1000$. For $n \geq 10$, $(n + 1)^3/n^3 \leq 1.1^3 < 2$, so if $2^n > n^3$ then $2^{n+1} > 2n^3 > (n + 1)^3$. So N = 10. (2 fails because n = 9 is a counterexample.)
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(C)** 5039	***Hint:*** *$k \cdot k! = (k + 1)! - k!$. Prove $\Sigma\, k \cdot k! = (n + 1)! - 1$ by induction.* The sum telescopes: $(2! - 1!) + (3! - 2!) + \ldots + (7! - 6!) = 7! - 1 = 5039$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(D)** 1/8	***Hint:*** *Count solutions of $a + b + c = 10$ with each value from 1 to 6 (the coefficient of x^{10} in $(x + x^2 + \ldots + x^6)^3$).* Solutions with a, b, c ≥ 1: C(9, 2) = 36. Remove those with a value ≥ 7: each such case leaves C(3, 2) = 3, for 3 × 3 = 9. Favourable = 27, so P = 27/216 = 1/8.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(B)** 1	***Hint:*** *Group the terms in blocks of four.* Each block of four consecutive powers adds to $1 + i - 1 - i = 0$. There are 101 terms: 25 blocks (i^0 to i^{99}) plus $i^{100} = 1$. Sum = 1. (As a GP: $(1 - i^{101})/(1 - i) = (1 - i)/(1 - i) = 1$.)
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(C)** $(\sqrt{6} - \sqrt{2})/4$	***Hint:*** *The distance is $1 \times \sin 15°$.* The foot of the perpendicular, O and (1, 0) form a right triangle with hypotenuse 1 and angle 15° at O. Distance = $\sin 15° = \sin(45° - 30°) = (\sqrt{6} - \sqrt{2})/4$. ($2 - \sqrt{3}$ is $\tan 15°$.)
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(B)** 89	***Hint:*** *The last move is a 1-step or a 2-step, so $w_n = w_{n-1} + w_{n-2}$.* $w_1 = 1$, $w_2 = 2$, then 3, 5, 8, 13, 21, 34, 55, 89. So $w_{10} = 89$.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-3

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. Let $z = 3 + 4i$. What is the value of $z/\bar{z} + \bar{z}/z$?

(A) $-14/25$ **(B)** $14/25$ **(C)** 2 **(D)** $-7/25$

Q2. Complex numbers z_1 and z_2 satisfy $|z_1| = 3$, $|z_2| = 4$ and $|z_1 + z_2| = 5$. What is $|z_1 - z_2|$?

(A) 1 **(B)** 5 **(C)** 7 **(D)** $\sqrt{7}$

Q3. What is the principal argument of $(1 + i)^2/(1 - i)$?

(A) $45°$ **(B)** $135°$ **(C)** $90°$ **(D)** $-135°$

Part 2 — Algebra (Syllabus Chapter 4)

Q4. Three numbers in AP have sum 24 and product 440. What is the largest of them?

(A) 10 **(B)** 12 **(C)** 11 **(D)** 13

Q5. Three numbers in GP have sum 21 and product 216. What is the sum of their squares?

(A) 441 **(B)** 189 **(C)** 147 **(D)** 216

Q6. What is the remainder when 9^{10} is divided by 64?

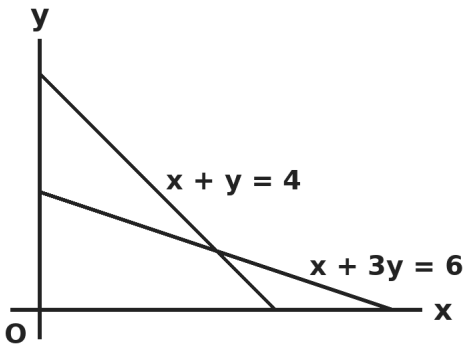
(A) 1 **(B)** 33 **(C)** 9 **(D)** 17

Q7. What is $1^2 + 3^2 + 5^2 + \ldots + 19^2$?

(A) 1540 **(B)** 385 **(C)** 1300 **(D)** 1330

Q8. The arithmetic mean of two positive numbers is 25 and their geometric mean is 20. What is the difference between the numbers?

(A) 10 **(B)** 40 **(C)** 30 **(D)** 15



Q9. x and y satisfy x + y ≤ 4, x + 3y ≤ 6, x ≥ 0 and y ≥ 0. What is the greatest value of 2x + 5y?

(A) 10 **(B)** 11 **(C)** 8 **(D)** 15

Part 3 — Geometry (Syllabus Chapter 5)

Q10. Two lines through the origin each make an angle of 45° with the line y = 2x. What is the product of their slopes?

(A) 1 **(B)** −3 **(C)** −1 **(D)** 1/3

Q11. In what ratio does the line 3x + y = 9 divide the segment joining (1, 3) and (2, 7)?

(A) 4 : 3 internally **(B)** 3 : 4 externally

(C) 1 : 2 internally **(D)** 3 : 4 internally

Q12. What is the area of the region |x| + |y| ≤ 4?

(A) 16 **(B)** 64 **(C)** 8 **(D)** 32

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of tan 20° × tan 40° × tan 80°?

(A) √3 **(B)** 1/√3 **(C)** 1 **(D)** 3

Q14. If tan θ = 3/4, what is cos 4θ?

(A) 527/625 **(B)** −527/625 **(C)** 7/25 **(D)** −336/625

Q15. What is the value of tan 9° − tan 27° − tan 63° + tan 81°?

(A) 0 **(B)** 2 **(C)** 4 **(D)** √3

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. In how many arrangements of the letters of EQUATION are all the vowels together?

(A) 2880 **(B)** 720 **(C)** 14400 **(D)** 4320

Q17. Eight friends sit round a circular table. In how many arrangements do Diya and Kunal not sit next to each other?

(A) 5040 **(B)** 1440 **(C)** 4320 **(D)** 3600

Q18. A fair die is thrown 6 times. What is the probability that an even number shows exactly 3 times?

(A) 5/16 **(B)** 1/2 **(C)** 1/8 **(D)** 3/8

Q19. Group A has 10 values with mean 20 and standard deviation 2. Group B has 10 values with mean 30 and standard deviation 4. What is the standard deviation of all 20 values together?

(A) 3 **(B)** $\sqrt{35}$ **(C)** $\sqrt{10}$ **(D)** 5

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. Bank A pays 10% per annum compounded half-yearly. Bank B pays 10.2% per annum compounded yearly. Ananya deposits ₹40,000 for one year. Which bank pays more, and by how much?

(A) Bank B, by ₹20 **(B)** Bank A, by ₹100

(C) Bank A, by ₹20 **(D)** Both pay the same

Q21. Gaurav repays a loan in three equal yearly instalments of ₹13,310 each, the first one year after borrowing. Interest is 10% per annum compounded yearly. How much did he borrow?

(A) ₹39,930 **(B)** ₹36,300 **(C)** ₹30,000 **(D)** ₹33,100

Q22. Pipe P fills a tank in 20 minutes and pipe Q in 30 minutes. Pipe R empties the full tank in 15 minutes. All three are opened together on the empty tank, and R is closed after 10 minutes. How long does the tank take to fill from the start?

(A) 60 min **(B)** 12 min **(C)** 16 min **(D)** 20 min

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the least number of distinct integers that must be chosen from 1, 2, 3, … , 30 to be sure that two of the chosen numbers add up to 31?

(A) 16 **(B)** 15 **(C)** 31 **(D)** 17

Q24. What is the largest possible number of acute interior angles in a convex polygon?

(A) 4 **(B)** 3 **(C)** 2 **(D)** No limit

Q25. How many solutions does $a + b + c + d = 8$ have in non-negative integers with $a \geq 2$?

(A) 84 **(B)** 165 **(C)** 56 **(D)** 120

Part 8 — Mixed Problems (each joins two chapters)

Q26. In how many ways can a 2×8 board be covered by eight 1×2 dominoes?

(A) 21 **(B)** 55 **(C)** 34 **(D)** 64

Q27. What is $C(10, 0) + C(10, 2) + C(10, 4) + \ldots + C(10, 10)$?

(A) 512 **(B)** 1024 **(C)** 256 **(D)** 462

Q28. α and β are the roots of $z^2 - 2z + 2 = 0$. What is $\alpha^6 + \beta^6$?

(A) 16 **(B)** −16 **(C)** 0 **(D)** 2

Q29. A line passes through $(\sqrt{3}, 1)$, and the perpendicular from the origin to it makes an angle of $30°$ with the positive x-axis. How far is the line from the origin?

(A) 2 **(B)** $\sqrt{3}$ **(C)** 1 **(D)** 4

Q30. In a group, every two people are either friends or strangers. What is the smallest group size that always contains 3 mutual friends or 3 mutual strangers?

(A) 5 **(B)** 6 **(C)** 7 **(D)** 9

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 3

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 3

Quick answers

1 **A**	6 **D**	11 **D**	16 **A**	21 **D**	26 **C**
2 **B**	7 **D**	12 **D**	17 **D**	22 **D**	27 **A**
3 **B**	8 **C**	13 **A**	18 **A**	23 **A**	28 **C**
4 **C**	9 **B**	14 **B**	19 **B**	24 **B**	29 **A**
5 **B**	10 **C**	15 **C**	20 **C**	25 **A**	30 **B**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(A)** −14/25	***Hint:*** *$z/\bar{z} = z^2/\|z\|^2$, and the two terms are conjugates of each other.* $z/\bar{z} = z^2/(z\,\bar{z}) = (-7 + 24i)/25$. The second term is its conjugate, so the sum is twice the real part: $2 \times (-7/25) = -14/25$.
2	Part 1 · Ch1 · Hard	**(B)** 5	***Hint:*** *$\|z_1 + z_2\|^2 + \|z_1 - z_2\|^2 = 2(\|z_1\|^2 + \|z_2\|^2)$.* $\|z_1 - z_2\|^2 = 2(9 + 16) - 25 = 25$, so $\|z_1 - z_2\| = 5$. (The vectors are at right angles, so both diagonals of the rectangle are 5.)
3	Part 1 · Ch1 · Hard	**(B)** 135°	***Hint:*** *Simplify to the form $a + bi$ first.* $(1 + i)^2 = 2i$. $2i/(1 - i) = 2i(1 + i)/2 = i + i^2 = -1 + i$. This lies in the second quadrant with real and imaginary parts of equal size, so the argument is 135°.
4	Part 2 · Ch4 · Hard	**(C)** 11	***Hint:*** *Take the numbers as $8 - d$, 8 and $8 + d$.* The middle number is $24 \div 3 = 8$. Then $8(64 - d^2) = 440$, so $d^2 = 9$, $d = \pm 3$. The numbers are 5, 8, 11, and the largest is 11.
5	Part 2 · Ch4 · Hard	**(B)** 189	***Hint:*** *Take the numbers as a/r, a and ar. The product gives a.* $a^3 = 216$, so $a = 6$. Then $6/r + 6 + 6r = 21$ gives $2r^2 - 5r + 2 = 0$, $r = 2$ or $1/2$. The numbers are 3, 6, 12, and $9 + 36 + 144 = 189$.
6	Part 2 · Ch4 · Hard	**(D)** 17	***Hint:*** *Write $9 = 1 + 8$ and expand.* $9^{10} = (1 + 8)^{10} = 1 + 10 \times 8 + C(10, 2) \times 64 + \ldots$ Every term from the third on is a multiple of 64. So $9^{10} \equiv 1 + 80 = 81 \equiv 17 \pmod{64}$.
7	Part 2 · Ch4 · Hard	**(D)** 1330	***Hint:*** *Subtract the even squares $2^2 + 4^2 + \ldots + 20^2 = 4(1^2 + \ldots + 10^2)$ from $1^2 + \ldots + 20^2$.* $1^2 + \ldots + 20^2 = 20 \cdot 21 \cdot 41/6 = 2870$. Even squares: $4 \times 385 = 1540$. Odd squares: $2870 - 1540 = 1330$.
8	Part 2 · Ch4 · Hard	**(C)** 30	***Hint:*** *The numbers have sum 50 and product 400.* $a + b = 50$ and $ab = 400$. $(a - b)^2 = 2500 - 1600 = 900$, so $a - b = 30$. The numbers are 40 and 10.
9	Part 2 · Ch4 · Hard	**(B)** 11	***Hint:*** *The greatest value occurs at a corner of the region.* Corners: (0, 0), (4, 0), (0, 2) and the meeting point of $x + y = 4$ and $x + 3y = 6$, which is (3, 1). Values of $2x + 5y$: 0, 8, 10, 11. The greatest is 11.
10	Part 3 · Ch5 · Hard	**(C)** −1	***Hint:*** *Solve $\|(m - 2)/(1 + 2m)\| = 1$.* $(m - 2)/(1 + 2m) = 1$ gives $m = -3$. $(m - 2)/(1 + 2m) = -1$ gives $m = 1/3$. Product = −1, so the two lines are perpendicular.

Q	Part · level	Answer	Why?
11	Part 3 · Ch5 · Hard	**(D)** 3 : 4 internally	***Hint:*** *Put each point into 3x + y − 9. The ratio is minus the ratio of the values.* At (1, 3): 3 + 3 − 9 = −3. At (2, 7): 6 + 7 − 9 = 4. The signs differ, so the division is internal, in the ratio 3 : 4.
12	Part 3 · Ch5 · Hard	**(D)** 32	***Hint:*** *The region is a square with vertices on the axes.* The vertices are (±4, 0) and (0, ±4). It is a square whose diagonals have length 8, so its area is 8 × 8 ÷ 2 = 32.
13	Part 4 · Ch7 · Hard	**(A)** √3	***Hint:*** *tan θ tan(60° − θ) tan(60° + θ) = tan 3θ.* With θ = 20°: tan 20° tan 40° tan 80° = tan 60° = √3.
14	Part 4 · Ch7 · Hard	**(B)** −527/625	***Hint:*** *First find cos 2θ = (1 − tan²θ)/(1 + tan²θ).* cos 2θ = (1 − 9/16)/(1 + 9/16) = 7/25. cos 4θ = 2cos²2θ − 1 = 2 × 49/625 − 1 = −527/625.
15	Part 4 · Ch7 · Hard	**(C)** 4	***Hint:*** *Pair tan 9° with tan 81° and tan 27° with tan 63°, and use tan θ + cot θ = 2/sin 2θ.* tan 9° + tan 81° = tan 9° + cot 9° = 2/sin 18°. tan 27° + tan 63° = 2/sin 54° = 2/cos 36°. Using sin 18° = (√5 − 1)/4 and cos 36° = (√5 + 1)/4: 8/(√5 − 1) − 8/(√5 + 1) = 8 × 2/4 = 4.
16	Part 5 · Ch8 · Hard	**(A)** 2880	***Hint:*** *Treat the five vowels as one block, then arrange inside the block.* Vowels E, U, A, I, O form one block. Block + Q, T, N = 4 items: 4! = 24 ways. Inside the block: 5! = 120. Total 24 × 120 = 2880.
17	Part 5 · Ch8 · Hard	**(D)** 3600	***Hint:*** *Total circular arrangements minus those with Diya and Kunal together.* Total: 7! = 5040. Together: treat them as one unit, 6! ways, times 2 for their order: 1440. Not together: 5040 − 1440 = 3600.
18	Part 5 · Ch8 · Hard	**(A)** 5/16	***Hint:*** *Each throw is even with probability 1/2. Use the binomial formula.* P = C(6, 3)(1/2)³(1/2)³ = 20/64 = 5/16.
19	Part 5 · Ch8 · Hard	**(B)** √35	***Hint:*** *Find the mean of the squares in each group, then combine.* Combined mean = 25. Mean of squares: A gives 4 + 400 = 404, B gives 16 + 900 = 916. Combined mean of squares = (404 + 916)/2 = 660. Variance = 660 − 625 = 35, so SD = √35.
20	Part 6 · Ch9 · Hard	**(C)** Bank A, by ₹20	***Hint:*** *Bank A multiplies the money by 1.05 twice.* Bank A: 40,000 × 1.05² = 40,000 × 1.1025 = ₹44,100. Bank B: 40,000 × 1.102 = ₹44,080. Bank A pays ₹20 more (its effective rate is 10.25%).
21	Part 6 · Ch9 · Hard	**(D)** ₹33,100	***Hint:*** *Add the present values 13,310/1.1 + 13,310/1.21 + 13,310/1.331.* Present values: 12,100 + 11,000 + 10,000 = ₹33,100. (₹39,930 ignores interest.)
22	Part 6 · Ch9 · Hard	**(D)** 20 min	***Hint:*** *Find the part filled in the first 10 minutes, then the rate of P and Q alone.* With all three: 1/20 + 1/30 − 1/15 = 1/60 tank per minute, so 10 minutes fill 1/6. P and Q alone fill 1/20 + 1/30 = 1/12 per minute, so the remaining 5/6 takes 10 minutes. Total 20 minutes.
23	Part 7 · Ch10 · Hard	**(A)** 16	***Hint:*** *Group the numbers into pairs that add up to 31.* The pairs (1, 30), (2, 29), … , (15, 16) make 15 boxes. With 16 numbers, two fall in the same pair and add to 31. The 15 numbers 1 to 15 have no such pair, so 15 is not enough. Answer: 16.

Q	Part · level	Answer	Why?
24	Part 7 · Ch10 · Hard	**(B)** 3	***Hint:*** *Think about the exterior angles, which add up to 360°.* An acute interior angle has an exterior angle greater than 90°. Four of them would make the exterior angles exceed 360°, which is impossible. A triangle with three acute angles shows that 3 can happen.
25	Part 7 · Ch10 · Hard	**(A)** 84	***Hint:*** *Put $a = 2 + a'$ and count solutions of $a' + b + c + d = 6$.* With $a' = a - 2 \geq 0$: $a' + b + c + d = 6$ has $C(6 + 3, 3) = C(9, 3) = 84$ solutions.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(C)** 34	***Hint:*** *Look at the left end: one vertical domino, or two horizontal ones stacked.* $t_n = t_{n-1} + t_{n-2}$, with $t_1 = 1$ and $t_2 = 2$. Then 3, 5, 8, 13, 21, 34. So $t_8 = 34$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(A)** 512	***Hint:*** *Add the expansions of $(1 + 1)^{10}$ and $(1 - 1)^{10}$.* $(1 + 1)^{10} + (1 - 1)^{10} = 2[C(10, 0) + C(10, 2) + \ldots]$. So the sum is $1024 \div 2 = 512$.
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(C)** 0	***Hint:*** *The roots are $1 \pm i$, and $(1 \pm i)^2 = \pm 2i$.* $\alpha = 1 + i$, $\beta = 1 - i$. $\alpha^6 = (2i)^3 = -8i$ and $\beta^6 = (-2i)^3 = 8i$. Sum = 0.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(A)** 2	***Hint:*** *The line is $x \cos 30° + y \sin 30° = p$.* Putting $(\sqrt{3}, 1)$ into $x \cos 30° + y \sin 30° = p$: $\sqrt{3} \times \sqrt{3}/2 + 1 \times 1/2 = 3/2 + 1/2 = 2$. So $p = 2$.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(B)** 6	***Hint:*** *Pick one person. Among the others, at least 3 are friends or at least 3 are strangers.* With 6 people, one person P has at least 3 friends or 3 strangers among the other 5 (pigeonhole). Say 3 friends: if any two of them are friends, they and P form a triangle of friends; if not, those 3 are mutual strangers. With 5 people, a pentagon of friendships (friends with neighbours only) has neither, so 5 is not enough.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-4

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. What is the value of $i^{57} + 1/i^{125} + i^{-27}$?

(A) 0 **(B)** −i **(C)** i **(D)** 1

Q2. What is the modulus of $(1 + i)^6(3 - 4i)^2/(2 + i)$?

(A) 200 **(B)** $40\sqrt{5}$ **(C)** 40 **(D)** $8\sqrt{5}$

Q3. A complex number z satisfies $|z - (3 + 4i)| = 2$. What is the least possible value of $|z|$?

(A) 5 **(B)** 7 **(C)** 3 **(D)** 1

Part 2 — Algebra (Syllabus Chapter 4)

Q4. What is the sum of all three-digit numbers that are divisible by 7?

(A) 70336 **(B)** 70436 **(C)** 69342 **(D)** 71064

Q5. The sums of the first n terms of two APs are in the ratio $(7n + 1) : (4n + 27)$ for every n. What is the ratio of their 11th terms?

(A) 4 : 3 **(B)** 78 : 71 **(C)** 7 : 4 **(D)** 3 : 2

Q6. Write 0.1232323… (with 23 repeating forever) as a fraction in lowest terms.

(A) 122/999 **(B)** 61/495 **(C)** 123/990 **(D)** 41/333

Q7. What is the coefficient of x^3 in the expansion of $(1 + 2x)^6(1 - x)^7$?

(A) −3 **(B)** 43 **(C)** −43 **(D)** 125

Q8. What is the least value of $(x^2 + 5)/\sqrt{(x^2 + 4)}$ for real x?

(A) 2 **(B)** 5/2 **(C)** $\sqrt{5}$ **(D)** 1

Q9. How many integers x satisfy both $|2x - 1| \le 7$ and $|x| > 1$?

(A) 8 **(B)** 6 **(C)** 3 **(D)** 5

Part 3 — Geometry (Syllabus Chapter 5)

Q10. A triangle has vertices (0, 0), (6, 0) and (0, 8). What is the distance between its centroid and its circumcentre?

(A) 5/3 **(B)** 5 **(C)** 10/3 **(D)** 4/3

Q11. What is the area of the parallelogram formed by the lines $y = x$, $y = x + 2$, $y = 2x$ and $y = 2x + 4$?

(A) 16 **(B)** 4 **(C)** 6 **(D)** 8

Q12. What is the mirror image of the origin in the line $3x + 4y = 25$?

(A) (3, 4) **(B)** (8, 6) **(C)** (−6, −8) **(D)** (6, 8)

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of $\sin 10° \times \sin 50° \times \sin 70°$?

(A) 1/8 **(B)** 1/4 **(C)** $\sqrt{3}/8$ **(D)** 1/16

Q14. If $\tan A = 1/2$ and $\tan B = 1/3$, what is $\tan(2A + B)$?

(A) 1 **(B)** 7/9 **(C)** −3 **(D)** 3

Q15. What is the value of $\tan 70° - \tan 20° - 2\tan 50°$?

(A) 1 **(B)** 0 **(C)** 2 **(D)** $\tan 50°$

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. How many rectangles on an 8 × 8 chessboard (made of the grid lines) are not squares?

(A) 1092 **(B)** 1296 **(C)** 204 **(D)** 1028

Q17. Four boys and three girls stand in a row. In how many ways can they stand if no two girls are next to each other?

(A) 720 **(B)** 5040 **(C)** 1440 **(D)** 144

Q18. In each of 5 independent trials, success has probability 1/3. What is the probability of at least 2 successes?

(A) 131/243 **(B)** 112/243 **(C)** 80/243 **(D)** 40/243

Q19. A variable x has mean 5 and variance 4. What are the mean and the standard deviation of y = 3 − 2x?

(A) −7 and −4 **(B)** 13 and 4 **(C)** −7 and 4 **(D)** −7 and 16

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. A shopkeeper in Pune buys an item for ₹1,000 plus 18% GST and sells it within Maharashtra for ₹1,416, including 18% GST. After input tax credit, how much CGST does he pay to the government?

(A) ₹36 **(B)** ₹18 **(C)** ₹108 **(D)** ₹90

Q21. Anil starts a business with ₹24,000. Four months later Prachi joins with ₹30,000. At the end of the year the profit is ₹22,000. What is Anil's share?

(A) ₹9,778 **(B)** ₹10,000 **(C)** ₹11,000 **(D)** ₹12,000

Q22. Machine A costs ₹20,000 to install and ₹30 per item to run. Machine B costs ₹35,000 to install and ₹25 per item. For how many items is machine B cheaper overall?

(A) Fewer than 3,000 **(B)** More than 3,000

(C) More than 1,400 **(D)** More than 7,000

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. A drawer has 2 red, 10 blue, 10 green and 10 black socks. Socks are taken out in the dark. What is the least number that must be taken out to be sure of getting 4 socks of the same colour?

(A) 12 **(B)** 13 **(C)** 16 **(D)** 10

Q24. How many positive divisors does 7200 have?

(A) 36 **(B)** 48 **(C)** 72 **(D)** 54

Q25. What is the least length L such that every list of L distinct numbers contains 5 numbers that are increasing, or 5 that are decreasing? (The 5 numbers need not be next to each other, but must be in their order in the list.)

(A) 16 **(B)** 25 **(C)** 10 **(D)** 17

Part 8 — Mixed Problems (each joins two chapters)

Q26. The sequence 1, 1, 2, 3, 5, 8, … has each term equal to the sum of the two before it. What is the sum of its first 10 terms?

(A) 144 **(B)** 89 **(C)** 143 **(D)** 232

Q27. What is the value of $C(6, 0) + 2 \cdot C(6, 1) + 4 \cdot C(6, 2) + 8 \cdot C(6, 3) + \ldots + 64 \cdot C(6, 6)$?

(A) 729 **(B)** 64 **(C)** 128 **(D)** 486

Q28. What is the value of $C(10, 1) - C(10, 3) + C(10, 5) - C(10, 7) + C(10, 9)$?

(A) 0 **(B)** −32 **(C)** 512 **(D)** 32

Q29. What is the acute angle between the lines $\sqrt{3}x + y = 0$ and $x + \sqrt{3}y = 0$?

(A) 60° **(B)** 45° **(C)** 30° **(D)** 90°

Q30. Triangles are formed by joining three vertices of a regular octagon. How many of these triangles have no side in common with the octagon?

(A) 24 **(B)** 16 **(C)** 32 **(D)** 56

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 4

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 4

Quick answers

1 **C**	6 **B**	11 **D**	16 **A**	21 **D**	26 **C**
2 **B**	7 **C**	12 **D**	17 **C**	22 **B**	27 **A**
3 **C**	8 **B**	13 **A**	18 **A**	23 **A**	28 **D**
4 **A**	9 **D**	14 **D**	19 **C**	24 **D**	29 **C**
5 **A**	10 **A**	15 **B**	20 **B**	25 **D**	30 **B**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(C)** i	***Hint:*** *Reduce each exponent modulo 4. Note 1/i = −i.* $i^{57} = i^1 = i$. $i^{125} = i$, so $1/i^{125} = 1/i = -i$. $-27 = -28 + 1$, so $i^{-27} = i$. Sum = i − i + i = i.
2	Part 1 · Ch1 · Hard	**(B)** 40√5	***Hint:*** *The modulus of a product or quotient is the product or quotient of the moduli.* $\lvert 1+i\rvert^6 = (\sqrt{2})^6 = 8$. $\lvert 3-4i\rvert^2 = 25$. $\lvert 2+i\rvert = \sqrt{5}$. Modulus = $8 \times 25/\sqrt{5} = 200/\sqrt{5} = 40\sqrt{5}$.
3	Part 1 · Ch1 · Hard	**(C)** 3	***Hint:*** *Use the triangle inequality: $\lvert 3+4i\rvert \le \lvert z\rvert + \lvert (3+4i) - z\rvert$.* $5 = \lvert 3+4i\rvert \le \lvert z\rvert + \lvert 3+4i-z\rvert = \lvert z\rvert + 2$, so $\lvert z\rvert \ge 3$. Equality holds for $z = (3/5)(3+4i) = 9/5 + (12/5)i$, which satisfies $\lvert z-(3+4i)\rvert = 2$. The least value is 3.
4	Part 2 · Ch4 · Hard	**(A)** 70336	***Hint:*** *The multiples run from 105 to 994. Count them first.* Number of terms: (994 − 105)/7 + 1 = 128. Sum = 128 × (105 + 994)/2 = 64 × 1099 = 70336.
5	Part 2 · Ch4 · Hard	**(A)** 4 : 3	***Hint:*** *The ratio of the 11th terms equals the ratio of the sums of the first 21 terms.* S_{21} = 21 × (11th term) for any AP. So the ratio of the 11th terms = $S_{21} : S'_{21} = (7 \times 21 + 1) : (4 \times 21 + 27) = 148 : 111 = 4 : 3$. (78 : 71 uses n = 11.)
6	Part 2 · Ch4 · Hard	**(B)** 61/495	***Hint:*** *0.1232323… = 1/10 + 0.0232323…, and 0.0232323… is a GP.* 0.0232323… = (23/1000)/(1 − 1/100) = 23/990. Total = 1/10 + 23/990 = 99/990 + 23/990 = 122/990 = 61/495.
7	Part 2 · Ch4 · Hard	**(C)** −43	***Hint:*** *Combine the terms up to x^3 of each factor.* $(1+2x)^6 = 1 + 12x + 60x^2 + 160x^3 + \ldots$ and $(1-x)^7 = 1 - 7x + 21x^2 - 35x^3 + \ldots$ The x^3 coefficient is 160 + 60(−7) + 12(21) + 1(−35) = 160 − 420 + 252 − 35 = −43.
8	Part 2 · Ch4 · Hard	**(B)** 5/2	***Hint:*** *Write it as $t + 1/t$ with $t = \sqrt{(x^2+4)} \ge 2$. Can AM ≥ GM be reached?* With $t = \sqrt{(x^2+4)}$, the expression is $(t^2+1)/t = t + 1/t$. AM ≥ GM would give 2, but that needs t = 1, which is impossible since t ≥ 2. On t ≥ 2 the value t + 1/t grows with t, so the least value is at t = 2 (x = 0): 2 + 1/2 = 5/2.
9	Part 2 · Ch4 · Hard	**(D)** 5	***Hint:*** *Solve $\lvert 2x-1\rvert \le 7$ first, then remove the integers with $\lvert x\rvert \le 1$.* $\lvert 2x-1\rvert \le 7$ gives $-3 \le x \le 4$: eight integers. Removing −1, 0, 1 leaves −3, −2, 2, 3, 4: five integers.

Q	Part · level	Answer	Why?
10	Part 3 · Ch5 · Hard	**(A)** 5/3	***Hint:*** *In a right triangle the circumcentre is the midpoint of the hypotenuse.* Centroid = (2, 8/3). Circumcentre = midpoint of (6, 0) and (0, 8) = (3, 4). Distance = $\sqrt{(1 + (4/3)^2)} = \sqrt{(25/9)} = 5/3$.
11	Part 3 · Ch5 · Hard	**(D)** 8	***Hint:*** *Find the four corners, or use area = $\lvert c_1 - c_2 \rvert \lvert d_1 - d_2 \rvert \div \lvert m_1 - m_2 \rvert$.* Corners: (0, 0), (2, 4), (−2, 0) and (−4, −4). By the shoelace formula, area = ½|0 + 8 + 8 + 0| = 8. The formula gives 2 × 4 ÷ |1 − 2| = 8 as well.
12	Part 3 · Ch5 · Hard	**(D)** (6, 8)	***Hint:*** *The foot of the perpendicular from O lies along the direction (3, 4).* The foot is t(3, 4) with 25t = 25, so (3, 4), at distance 5. The image is twice as far along the same direction: (6, 8).
13	Part 4 · Ch7 · Hard	**(A)** 1/8	***Hint:*** *$\sin\theta \sin(60° - \theta) \sin(60° + \theta) = \frac{1}{4} \sin 3\theta$.* With θ = 10°: sin 10° sin 50° sin 70° = ¼ sin 30° = ¼ × ½ = 1/8.
14	Part 4 · Ch7 · Hard	**(D)** 3	***Hint:*** *First find tan 2A.* tan 2A = (2 × ½)/(1 − ¼) = 4/3. tan(2A + B) = (4/3 + 1/3)/(1 − 4/9) = (5/3)/(5/9) = 3.
15	Part 4 · Ch7 · Hard	**(B)** 0	***Hint:*** *50° = 70° − 20°, and tan 70° × tan 20° = 1.* tan 50° = tan(70° − 20°) = (tan 70° − tan 20°)/(1 + tan 70° tan 20°). Since tan 70° = cot 20°, the product tan 70° tan 20° = 1, so tan 70° − tan 20° = 2 tan 50°. The value is 0.
16	Part 5 · Ch8 · Hard	**(A)** 1092	***Hint:*** *Rectangles: choose 2 of the 9 vertical lines and 2 of the 9 horizontal lines. Then subtract the squares.* Rectangles = $C(9, 2)^2 = 36^2 = 1296$. Squares = $1^2 + 2^2 + \ldots + 8^2 = 204$. Non-square rectangles = 1296 − 204 = 1092.
17	Part 5 · Ch8 · Hard	**(C)** 1440	***Hint:*** *Seat the boys first. The girls go into the gaps between and around them.* Boys: 4! = 24 ways. They leave 5 gaps; choose 3 for the girls and arrange the girls: C(5, 3) × 3! = 60. Total 24 × 60 = 1440.
18	Part 5 · Ch8 · Hard	**(A)** 131/243	***Hint:*** *Subtract P(0) and P(1) from 1.* $P(0) = (2/3)^5 = 32/243$. $P(1) = 5 \times (1/3)(2/3)^4 = 80/243$. P(at least 2) = 1 − 112/243 = 131/243.
19	Part 5 · Ch8 · Hard	**(C)** −7 and 4	***Hint:*** *Adding a constant shifts the mean only; multiplying by −2 multiplies the SD by 2.* Mean of y = 3 − 2 × 5 = −7. Variance of $y = (-2)^2 \times 4 = 16$, so SD = 4 (an SD is never negative; 16 is the variance).
20	Part 6 · Ch9 · Hard	**(B)** ₹18	***Hint:*** *Remove the GST from the selling price first, then subtract the input GST.* Selling price before tax = 1,416 ÷ 1.18 = ₹1,200, so output GST = ₹216. Input GST = ₹180. Net GST = ₹36, split equally into CGST and SGST: CGST = ₹18. (₹108 forgets the input credit.)
21	Part 6 · Ch9 · Hard	**(D)** ₹12,000	***Hint:*** *Shares are in the ratio of money × months.* Anil: 24,000 × 12 = 2,88,000. Prachi: 30,000 × 8 = 2,40,000. Ratio 6 : 5. Anil gets 22,000 × 6/11 = ₹12,000.
22	Part 6 · Ch9 · Hard	**(B)** More than 3,000	***Hint:*** *Solve 35,000 + 25n < 20,000 + 30n.* 35,000 + 25n < 20,000 + 30n gives 15,000 < 5n, so n > 3,000. B is cheaper for more than 3,000 items.

Q	Part · level	Answer	Why?
23	Part 7 · Ch10 · Hard	**(A)** 12	***Hint:*** *Find the largest number you can take with at most 3 of each colour; there are only 2 red socks.* At most 2 red + 3 blue + 3 green + 3 black = 11 socks can be taken without 4 of one colour. The 12th sock must give 4 blue, green or black. (13 assumes 3 red socks are available.)
24	Part 7 · Ch10 · Hard	**(D)** 54	***Hint:*** *Write 7200 as a product of prime powers.* $7200 = 2^5 \times 3^2 \times 5^2$. A divisor is $2^a3^b5^c$ with a = 0 to 5, b = 0 to 2, c = 0 to 2: $6 \times 3 \times 3 = 54$.
25	Part 7 · Ch10 · Hard	**(D)** 17	***Hint:*** *Label each number with the length of the longest increasing subsequence ending at it.* Label each term by the longest increasing subsequence ending there. If none reaches 5, labels are 1 to 4, so with 17 terms some label is shared by 5 terms (pigeonhole), and those 5 must be decreasing. A list of 16 made of 4 decreasing blocks of 4, arranged in increasing order of blocks, has no monotone subsequence of 5. So L = 17.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(C)** 143	***Hint:*** *Prove by induction that $F_1 + F_2 + \ldots + F_n = F_{n+2} - 1$.* The identity holds for n = 1 (1 = 2 − 1), and adding F_{n+1} to both sides gives $F_{n+2} - 1 + F_{n+1} = F_{n+3} - 1$. So the sum is $F_{12} - 1 = 144 - 1 = 143$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(A)** 729	***Hint:*** *Compare with the binomial expansion of $(1 + 2)^6$.* $\Sigma\, C(6, k)2^k = (1 + 2)^6 = 3^6 = 729$. (It also counts the ways to give each of 6 children one of 3 colours, by first choosing who gets a non-first colour.)
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(D)** 32	***Hint:*** *These are the imaginary parts of the terms of $(1 + i)^{10}$.* $(1 + i)^{10} = \Sigma\, C(10, k)i^k$. The odd k give imaginary terms with signs +, −, +, −, +, so the value is $\mathrm{Im}(1 + i)^{10}$. Since $(1 + i)^{10} = (2i)^5 = 32i$, the value is 32. (Check: 10 − 120 + 252 − 120 + 10 = 32.)
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(C)** 30°	***Hint:*** *Find the inclination of each line from its slope.* Slopes $-\sqrt{3}$ and $-1/\sqrt{3}$ give inclinations 120° and 150°. The angle between them is 30°. (Formula: $\tan\theta = |(-\sqrt{3} + 1/\sqrt{3})/(1 + 1)| = 1/\sqrt{3}$.)
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(B)** 16	***Hint:*** *Count triangles with exactly one shared side and with two shared sides, then subtract.* Total C(8, 3) = 56. Two shared sides (three consecutive vertices): 8. Exactly one shared side: for each of the 8 sides, the third vertex avoids the side and its two neighbours: 8 − 4 = 4 choices, so 32. No shared side: 56 − 8 − 32 = 16.

ARYABHATTA MATHS OLYMPIAD

Level 2

PRACTICE PAPER-5

CLASS 11

Total Questions	30 multiple-choice questions
Level	Level-2 (final round): every question is Hard
Time	45 minutes
Arrangement	Parts by chapter; the last Part has mixed problems joining two chapters
Negative Marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Algebra · Geometry · Trigonometry · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning

Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

For teachers

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2, 3 and 6 are not part of the Class 11 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

Part 1 — Number System (Syllabus Chapter 1)

Q1. For how many integers n from 1 to 100 is $(1 + i)^n$ a real number?

(A) 25 **(B)** 50 **(C)** 24 **(D)** 26

Q2. Let $z = (3 + 4i)/5$. What is $(z - 1)/(z + 1)$?

(A) $i/2$ **(B)** $2i$ **(C)** $-i/2$ **(D)** $(1 + 2i)/5$

Q3. How many complex numbers z satisfy $z^2 + |z| = 0$?

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Part 2 — Algebra (Syllabus Chapter 4)

Q4. Some arithmetic means are inserted between 7 and 71. The 5th of these means is 27. How many means are inserted?

(A) 16 **(B)** 14 **(C)** 17 **(D)** 15

Q5. In a GP, the sum of the first 3 terms and the sum of the first 6 terms are in the ratio 125 : 152. What is the common ratio?

(A) 3/5 **(B)** 5/3 **(C)** 27/125 **(D)** 2/5

Q6. What is the value of $1{\cdot}2{\cdot}3 + 2{\cdot}3{\cdot}4 + 3{\cdot}4{\cdot}5 + \ldots + 10{\cdot}11{\cdot}12$?

(A) 3080 **(B)** 4004 **(C)** 5148 **(D)** 4290

Q7. What is the value of $1/(1{\cdot}3) + 1/(3{\cdot}5) + 1/(5{\cdot}7) + \ldots + 1/(19{\cdot}21)$?

(A) 20/21 **(B)** 10/21 **(C)** 1/2 **(D)** 11/21

Q8. What is the value of $(\sqrt{3} + 1)^6 + (\sqrt{3} - 1)^6$?

(A) 208 **(B)** 416 **(C)** 832 **(D)** 448

Q9. What is the area of the region where $x \geq 0$, $y \geq 0$, $x + 2y \leq 8$ and $3x + y \leq 9$?

(A) 17/2 **(B)** 12 **(C)** 16 **(D)** 15/2

Part 3 — Geometry (Syllabus Chapter 5)

Q10. A line through (3, 4) has x-intercept a and y-intercept 14 − a. Which values can a take?

(A) 7 only **(B)** 6 or 7 **(C)** 6 or 8 **(D)** 4 or 10

Q11. Which line bisects the angle between x + 2y = 3 and 2x + y = 3 that contains the origin?

(A) x + y = 2 **(B)** x − y = 0 **(C)** x + y = 0 **(D)** x − y = 2

Q12. What is the foot of the perpendicular from (−1, 3) to the line 3x − 4y − 16 = 0?

(A) (68/25, −49/25) **(B)** (4, −1)

(C) (−49/25, 68/25) **(D)** (2, −5/2)

Part 4 — Trigonometry (Syllabus Chapter 7)

Q13. What is the value of sin 105° + cos 105°?

(A) 1/√2 **(B)** −1/√2 **(C)** √3/2 **(D)** (√6 + √2)/2

Q14. What is the value of sin 20° × sin 40° × sin 80°?

(A) 1/8 **(B)** √3/8 **(C)** √3/4 **(D)** 3/8

Q15. If tan(A + B) = 3 and tan(A − B) = 2, what is tan 2A?

(A) 1 **(B)** 5 **(C)** −1 **(D)** 1/7

Part 5 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

Q16. A convex polygon has 54 diagonals. How many sides does it have?

(A) 11 **(B)** 9 **(C)** 18 **(D)** 12

Q17. In how many ways can 5 different balls be put into 3 different boxes so that no box is empty?

(A) 150 **(B)** 243 **(C)** 60 **(D)** 90

Q18. X is binomial with n = 6. If P(X = 2) = P(X = 3), what is p?

(A) 3/7 **(B)** 1/2 **(C)** 4/7 **(D)** 2/5

Q19. Ten observations have mean 12 and variance 9. Later it was found that one value, 15, had been read as 10. What is the correct variance?

(A) 9 **(B)** 10.5 **(C)** 9.25 **(D)** 8.75

Part 6 — Applied Mathematics (Syllabus Chapter 9)

Q20. A machine bought for ₹50,000 loses 20% of its value in the first year and 10% of its value in each later year. What is it worth after 3 years?

(A) ₹30,000 **(B)** ₹35,000 **(C)** ₹32,400 **(D)** ₹36,450

Q21. Ruhi borrows ₹20,100 and repays it in two equal monthly instalments at 1% interest per month (compounded monthly). The first instalment is paid one month after borrowing. What is each instalment?

(A) ₹10,151 **(B)** ₹10,050 **(C)** ₹10,251 **(D)** ₹10,201

Q22. A workshop makes x chairs and y stools a day, with $x + y \le 10$ and $2x + y \le 16$, and $x, y \ge 0$. Profit is ₹40 per chair and ₹30 per stool. What is the greatest daily profit?

(A) ₹320 **(B)** ₹360 **(C)** ₹300 **(D)** ₹400

Part 7 — Mathematical Reasoning (Syllabus Chapter 10)

Q23. What is the least n such that among any n integers there are always three whose sum is divisible by 3?

(A) 4 **(B)** 6 **(C)** 5 **(D)** 7

Q24. What are the last two digits of 7^{100}?

(A) 49 **(B)** 43 **(C)** 07 **(D)** 01

Q25. How many zeros are there at the end of 100! when it is written out in full?

(A) 20 **(B)** 24 **(C)** 25 **(D)** 11

Part 8 — Mixed Problems (each joins two chapters)

Q26. A sequence has $u_1 = 2$ and $u_{n+1} = u_n + 2n$ for $n \geq 1$. What is u_{20}?

(A) 380 **(B)** 402 **(C)** 422 **(D)** 382

Q27. What is the value of $C(5, 0)^2 + C(5, 1)^2 + C(5, 2)^2 + \ldots + C(5, 5)^2$?

(A) 1024 **(B)** 126 **(C)** 252 **(D)** 32

Q28. What is the value of $1 \cdot i + 2 \cdot i^2 + 3 \cdot i^3 + \ldots + 8 \cdot i^8$?

(A) 0 **(B)** $4 - 4i$ **(C)** $-4 + 4i$ **(D)** $4 + 4i$

Q29. A line passes through (2, 3) and makes an angle of 75° with the positive x-axis. Where does it cut the y-axis?

(A) $(0, 1 + 2\sqrt{3})$ **(B)** $(0, 7 + 2\sqrt{3})$ **(C)** $(0, -1 - 2\sqrt{3})$ **(D)** $(0, -1 + 2\sqrt{3})$

Q30. Four people are chosen at random. Each is equally likely to be born in any of the 12 months, independently. What is the probability that at least two share a birth month?

(A) 55/96 **(B)** 1/3 **(C)** 11/24 **(D)** 41/96

Answer Sheet AMO Level 2 · Class 11 · Practice Paper 5

Name: **School:** **Roll No.:**

Shade or tick one circle for each question.

1.	A	B	C	D	11.	A	B	C	D	21.	A	B	C	D
2.	A	B	C	D	12.	A	B	C	D	22.	A	B	C	D
3.	A	B	C	D	13.	A	B	C	D	23.	A	B	C	D
4.	A	B	C	D	14.	A	B	C	D	24.	A	B	C	D
5.	A	B	C	D	15.	A	B	C	D	25.	A	B	C	D
6.	A	B	C	D	16.	A	B	C	D	26.	A	B	C	D
7.	A	B	C	D	17.	A	B	C	D	27.	A	B	C	D
8.	A	B	C	D	18.	A	B	C	D	28.	A	B	C	D
9.	A	B	C	D	19.	A	B	C	D	29.	A	B	C	D
10.	A	B	C	D	20.	A	B	C	D	30.	A	B	C	D

For the parent or teacher

Correct: _____ / 30

Answer Key and Solutions AMO Level 2 · Class 11 · Practice Paper 5

Quick answers

1 **A**	6 **D**	11 **B**	16 **D**	21 **D**	26 **D**
2 **A**	7 **B**	12 **A**	17 **A**	22 **B**	27 **C**
3 **C**	8 **B**	13 **A**	18 **A**	23 **C**	28 **B**
4 **D**	9 **A**	14 **B**	19 **C**	24 **D**	29 **C**
5 **A**	10 **B**	15 **C**	20 **C**	25 **B**	30 **D**

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Hard	**(A)** 25	***Hint:*** *$(1 + i)^2 = 2i$. Consider even and odd n separately.* For even $n = 2m$, $(1 + i)^n = (2i)^m = 2^m i^m$, which is real exactly when m is even, that is n is a multiple of 4. For odd n, $(1 + i)^n = (1 + i)(2i)^m$ has both parts non-zero. So $n = 4, 8, \ldots, 100$: 25 values.
2	Part 1 · Ch1 · Hard	**(A)** i/2	***Hint:*** *Simplify the top and bottom first: $z - 1 = (-2 + 4i)/5$ and $z + 1 = (8 + 4i)/5$.* $(z - 1)/(z + 1) = (-2 + 4i)/(8 + 4i) = (-1 + 2i)/(4 + 2i) = (-1 + 2i)(4 - 2i)/20 = (-4 + 2i + 8i + 4)/20 = 10i/20 = i/2$. (It is purely imaginary because $\lvert z \rvert = 1$.)
3	Part 1 · Ch1 · Hard	**(C)** 3	***Hint:*** *Write $z = a + bi$. The imaginary part of z^2 must be 0, so $a = 0$ or $b = 0$.* $\mathrm{Im}(z^2) = 2ab = 0$. If $b = 0$: $a^2 + \lvert a \rvert = 0$ gives $a = 0$. If $a = 0$: $-b^2 + \lvert b \rvert = 0$ gives $b = 0, 1$ or -1. Solutions: 0, i, −i: three.
4	Part 2 · Ch4 · Hard	**(D)** 15	***Hint:*** *The 5th mean is the 6th term of the AP that starts with 7.* $7 + 5d = 27$ gives $d = 4$. If n means are inserted, 71 is term $n + 2$: $7 + (n + 1)4 = 71$, so $n + 1 = 16$ and $n = 15$.
5	Part 2 · Ch4 · Hard	**(A)** 3/5	***Hint:*** *$S_6 = S_3(1 + r^3)$.* $S_3 : S_6 = 1 : (1 + r^3) = 125 : 152$, so $1 + r^3 = 152/125$ and $r^3 = 27/125$. $r = 3/5$.
6	Part 2 · Ch4 · Hard	**(D)** 4290	***Hint:*** *Show that $\Sigma\, k(k + 1)(k + 2) = n(n + 1)(n + 2)(n + 3)/4$.* Using the formula with $n = 10$: $10 \times 11 \times 12 \times 13/4 = 4290$. (The formula telescopes: $k(k + 1)(k + 2) = [k(k + 1)(k + 2)(k + 3) - (k - 1)k(k + 1)(k + 2)]/4$.)
7	Part 2 · Ch4 · Hard	**(B)** 10/21	***Hint:*** *$1/((2k - 1)(2k + 1)) = \frac{1}{2}[1/(2k - 1) - 1/(2k + 1)]$.* The sum telescopes to $\frac{1}{2}(1 - 1/21) = \frac{1}{2} \times 20/21 = 10/21$. (20/21 forgets the ½.)
8	Part 2 · Ch4 · Hard	**(B)** 416	***Hint:*** *In the sum, the terms with odd powers of 1 cancel. Only even-k terms remain, doubled.* Sum $= 2[C(6, 0)(\sqrt{3})^6 + C(6, 2)(\sqrt{3})^4 + C(6, 4)(\sqrt{3})^2 + C(6, 6)] = 2[27 + 15 \times 9 + 15 \times 3 + 1] = 2 \times 208 = 416$.
9	Part 2 · Ch4 · Hard	**(A)** 17/2	***Hint:*** *Find the corner where the two lines meet, then use the shoelace formula.* The lines meet at (2, 3). Corners: (0, 0), (3, 0), (2, 3), (0, 4). Area $= \frac{1}{2}\lvert 0 + 3 \times 3 - 0 + 2 \times 4 - 0 + 0 \rvert = \frac{1}{2}(9 + 8) = 17/2$.

Q	Part · level	Answer	Why?
10	Part 3 · Ch5 · Hard	**(B)** 6 or 7	***Hint:*** *Put (3, 4) into x/a + y/(14 − a) = 1.* 3/a + 4/(14 − a) = 1 gives 3(14 − a) + 4a = a(14 − a), so a^2 − 13a + 42 = 0 and a = 6 or 7. The lines are x/6 + y/8 = 1 and x/7 + y/7 = 1.
11	Part 3 · Ch5 · Hard	**(B)** x − y = 0	***Hint:*** *At the origin both x + 2y − 3 and 2x + y − 3 are negative, so take the bisector with equal signs.* The bisectors are (x + 2y − 3)/√5 = ±(2x + y − 3)/√5. The origin makes both expressions −3, the same sign, so its angle is bisected by the "+" choice: x + 2y − 3 = 2x + y − 3, that is x − y = 0. (x + y = 2 bisects the other angle.)
12	Part 3 · Ch5 · Hard	**(A)** (68/25, −49/25)	***Hint:*** *Move from (−1, 3) along (3, −4) by t, where* $t = -(3x_0 - 4y_0 - 16)/25$. 3(−1) − 4(3) − 16 = −31, so t = 31/25. Foot = (−1 + 93/25, 3 − 124/25) = (68/25, −49/25). Check: 3 × 68/25 + 4 × 49/25 = 400/25 = 16.
13	Part 4 · Ch7 · Hard	**(A)** 1/√2	***Hint:*** *sin x + cos x = √2 sin(x + 45°).* sin 105° + cos 105° = √2 sin 150° = √2 × ½ = 1/√2. (Directly: (√6 + √2)/4 + (√2 − √6)/4 = √2/2.)
14	Part 4 · Ch7 · Hard	**(B)** √3/8	***Hint:*** *sin θ sin(60° − θ) sin(60° + θ) = ¼ sin 3θ.* With θ = 20°: sin 20° sin 40° sin 80° = ¼ sin 60° = ¼ × √3/2 = √3/8.
15	Part 4 · Ch7 · Hard	**(C)** −1	***Hint:*** *2A = (A + B) + (A − B).* tan 2A = (3 + 2)/(1 − 3 × 2) = 5/(−5) = −1. (1/7 is tan 2B.)
16	Part 5 · Ch8 · Hard	**(D)** 12	***Hint:*** *A polygon with n sides has C(n, 2) − n diagonals.* n(n − 1)/2 − n = n(n − 3)/2 = 54, so n^2 − 3n − 108 = 0 and n = 12.
17	Part 5 · Ch8 · Hard	**(A)** 150	***Hint:*** *Use inclusion–exclusion:* $3^5 - 3 \times 2^5 + 3 \times 1^5$. All ways: 3^5 = 243. Remove those missing a box: 3 × 2^5 = 96, then add back those missing two boxes: 3. 243 − 96 + 3 = 150.
18	Part 5 · Ch8 · Hard	**(A)** 3/7	***Hint:*** *C(6, 2)p^2q^4 = C(6, 3)p^3q^3.* $15p^2q^4 = 20p^3q^3$ gives 15q = 20p, so 3(1 − p) = 4p and p = 3/7. (p = 1/2 would need C(6, 2) = C(6, 3).)
19	Part 5 · Ch8 · Hard	**(C)** 9.25	***Hint:*** *Correct the sum and the sum of squares, then recompute.* Σx = 120 → 125, so the mean is 12.5. Σx^2 = 10(9 + 144) = 1530 → 1530 − 100 + 225 = 1655. Variance = 165.5 − 12.5^2 = 165.5 − 156.25 = 9.25.
20	Part 6 · Ch9 · Hard	**(C)** ₹32,400	***Hint:*** *Multiply by 0.8 once and by 0.9 twice.* 50,000 × 0.8 × 0.9 × 0.9 = ₹32,400. (Subtracting 40% of the original gives ₹30,000.)
21	Part 6 · Ch9 · Hard	**(D)** ₹10,201	***Hint:*** *The present values x/1.01 and x/1.0201 must add up to ₹20,100.* x(1/1.01 + 1/1.0201) = x × 2.01/1.0201 = 20,100, so x = 20,100 × 1.0201/2.01 = ₹10,201. Check: 10,201/1.01 + 10,201/1.0201 = 10,100 + 10,000.
22	Part 6 · Ch9 · Hard	**(B)** ₹360	***Hint:*** *Check the profit at each corner of the allowed region.* Corners: (0, 0), (8, 0), (6, 4), (0, 10). Profits: 0, 320, 240 + 120 = 360, 300. The greatest is ₹360 at 6 chairs and 4 stools.
23	Part 7 · Ch10 · Hard	**(C)** 5	***Hint:*** *Look at the remainders 0, 1, 2 modulo 3.* Among 5 integers, either all three remainders appear (one of each sums to 0 mod 3) or some remainder appears 3 times (pigeonhole), and three equal remainders also sum to 0 mod 3. The four numbers 0, 0, 1, 1 have no such triple, so 4 is not enough.

Q	Part · level	Answer	Why?
24	Part 7 · Ch10 · Hard	**(D)** 01	***Hint:*** *$7^4 = 2401$.* $7^4 = 2401 \equiv 1 \pmod{100}$. So $7^{100} = (7^4)^{25} \equiv 1 \pmod{100}$: the last two digits are 01.
25	Part 7 · Ch10 · Hard	**(B)** 24	***Hint:*** *Count the factors of 5; there are always enough 2s.* Multiples of 5 up to 100 give 20 fives, and multiples of 25 give 4 more. Total 24 factors of 5, so 24 trailing zeros.
26	Part 8 · Mixed · Ch4 + Ch10 · Hard	**(D)** 382	***Hint:*** *Guess a formula from the first few terms and prove it by induction.* $u_2 = 4$, $u_3 = 8$, $u_4 = 14$. The formula $u_n = n^2 - n + 2$ fits, and if it holds for n then $u_{n+1} = n^2 - n + 2 + 2n = (n + 1)^2 - (n + 1) + 2$. So $u_{20} = 400 - 20 + 2 = 382$.
27	Part 8 · Mixed · Ch4 + Ch8 · Hard	**(C)** 252	***Hint:*** *Choose 5 people from 5 men and 5 women: k women and 5 − k men.* $\Sigma\, C(5, k)\, C(5, 5 - k)$ counts 5-member groups from 10 people, so the sum is $C(10, 5) = 252$. ($32 = 2^5$ adds the coefficients without squaring them, and $1024 = 2^{10}$ squares that total instead of adding the squares.)
28	Part 8 · Mixed · Ch1 + Ch4 · Hard	**(B)** 4 − 4i	***Hint:*** *Group the terms in blocks of four: $k\, i^k$ for k = 1 to 4 and 5 to 8.* First block: $i - 2 - 3i + 4 = 2 - 2i$. Second block: $5i - 6 - 7i + 8 = 2 - 2i$. Total $4 - 4i$.
29	Part 8 · Mixed · Ch5 + Ch7 · Hard	**(C)** $(0, -1 - 2\sqrt{3})$	***Hint:*** *$\tan 75° = \tan(45° + 30°) = 2 + \sqrt{3}$.* Slope $= 2 + \sqrt{3}$. $y - 3 = (2 + \sqrt{3})(x - 2)$. At $x = 0$: $y = 3 - 2(2 + \sqrt{3}) = -1 - 2\sqrt{3}$.
30	Part 8 · Mixed · Ch8 + Ch10 · Hard	**(D)** 41/96	***Hint:*** *Find the probability that all four months are different.* All different: $(12 \times 11 \times 10 \times 9)/12^4 = 11880/20736 = 55/96$. At least two the same: $1 - 55/96 = 41/96$.